

Name: _____ / 25

Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

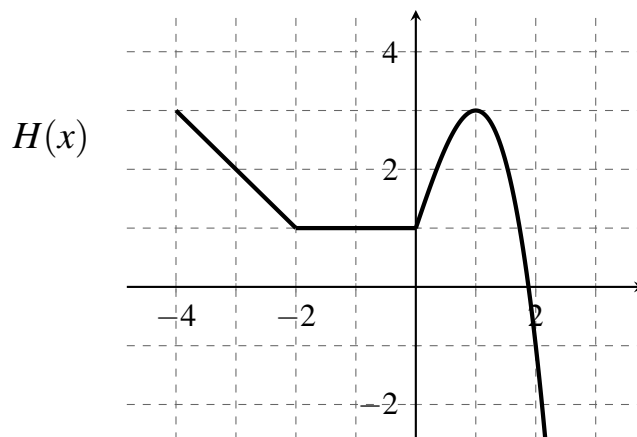
1. [10 points] The **derivative** of $f(x)$ is defined as follows:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

- a. Use the definition of the derivative (above) to find the derivative of the function $f(x) = \frac{5}{x+2}$. Use proper limit notation. No credit will be awarded for finding the derivative via other methods.

- b. Use your answer in part (a) to find the **slope** of the line tangent to $f(x) = \frac{5}{x+2}$ at the point $(3, 1)$. Simplify your answer.

2. [5 points] The function $H(x)$ is graphed below. Sketch the graph of $H'(x)$, the derivative of $H(x)$, on the same set of axes.



3. [10 points] Find $f'(x)$ for each function below. You may use any method you like, and you do not need to simplify your answer.

a. $f(x) = 8x^3 - 2\sqrt{x} + \frac{1}{x} - \pi^2$

b. $f(x) = (x+1)\cos(x)$

c. $f(x) = \frac{\sin(x)}{5x-4}$